CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION

Cambridge IGCSE™ & O Level Computer Science (0478 / 2210)

CHAPTER 1 (TOPICS 1.1 & 1.2) PAST EXAM QUESTION BANK (2023 – 2026 SERIES)

Syllabus Topics: 1.1 Number Systems & 1.2 Text, Sound, Images **Total Marks: 65 Marks**

TOPIC 1.1: NUMBER SYSTEMS — PAST PAPER EXAM QUESTIONS

Question 1 Cambridge 2210/12/M/J/24 Q1 [6 marks]

(a) Convert the 8-bit binary number `01011010` to denary. [1]

Answer:

(b) Convert the denary number **−45** to an 8-bit binary number using two's complement representation. [2]

Step-by-step working:

(c) Perform the binary addition of `01011010` and `00110111`. State whether an overflow error has occurred and explain your answer. [3]

Addition & Overflow explanation:

Question 2 Cambridge 2210/11/M/J/24 Q1 [6 marks]

(a) Convert the hexadecimal value **3B** into: [2]

(i) 8-bit Binary: _____ (ii) Denary: _____

(b) State two reasons why computer scientists use hexadecimal notation instead of binary. [2]

Reasons:

(c) Give two applications of hexadecimal in computer systems. [2]

Applications:

Question 3 Cambridge 2210/12/O/N/23 Q1 [5 marks]

An 8-bit register contains the binary value: `0 0 1 1 0 1 0 0` (Denary: 52).

(a) Perform a logical shift of 2 places to the left. Write the binary contents and new denary value. [2]

Binary: _____ Denary: _____

(b) Explain the arithmetic effect of a logical left shift and state what happens if bits are shifted out of the MSB. [3]

Explanation:

Question 4 Cambridge 0478/12/F/M/24 Q1 [5 marks]

(a) State the range of integer values that can be represented using an 8-bit signed two's complement binary register. [2]

Minimum value: _____ Maximum value: _____

(b) Convert the two's complement binary number `11001000` to denary. Show your working. [3]

Working & Denary value:

TOPIC 1.2: TEXT, SOUND AND IMAGES — PAST PAPER EXAM QUESTIONS

Question 5 Cambridge 2210/12/M/J/23 Q1 [6 marks]

(a) Explain the difference between the ASCII character set and the Unicode character set. [3]

ASCII vs Unicode:

(b) A text document contains **2,048 characters** encoded in 8-bit Extended ASCII. Calculate the document file size in **Kibibytes (KiB)**. Show your working. [3]

Calculation:

Question 6 Cambridge 2210/12/M/J/24 Q3 [7 marks]

(a) Explain how analogue sound waves are sampled to create a digital sound recording. [3]

Sampling explanation:

(b) Define the terms **sample rate** and **sample resolution (bit depth)**. [2]

Definitions:

(c) Describe the effect on sound file size and sound quality when sample rate is increased. [2]

Effects:

Question 7 Cambridge 2210/11/O/N/23 Q2 [4 marks]

A mono audio recording is made for **10 seconds** with a sample rate of **44,100 Hz** and a sample resolution of **16 bits**. Calculate the uncompressed file size in **Megabytes (MB)**. Show your full working. [4]

Working & Final Answer:

Question 8 Cambridge 0478/12/F/M/23 Q3 [7 marks]

(a) Define the terms **pixel**, **image resolution**, and **colour depth**. [3]

Definitions:

(b) A bitmap image has a resolution of **1000 × 800 pixels** and a colour depth of **24 bits**. Calculate the uncompressed image file size in **Mebibytes (MiB)**. Show your working. [4]

Calculation:

Question 9 Cambridge 2210/12/O/N/24 Q2 [6 marks]

(a) Compare **lossy compression** and **lossless compression**. State one file format suitable for each type. [4]

Comparison:

(b) State two benefits of compressing files before transmitting them over the Internet. [2]

Benefits:

Question 10 Cambridge 2210/11/M/J/23 Q4 [4 marks]

(a) Explain how **Run-Length Encoding (RLE)** is used to compress data. [2]

Explanation:

(b) The following row of pixels is to be compressed: `W W W W W  B B B  W W  G G G G`.
Show how this sequence is stored using RLE. [2]

RLE compressed format:
